Report

of Research Work conducted on

Biomedical Named Entity Recognition and Entity Linking with UMLS

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ABSTRACT

Biomedical Named Entity Recognition (Bio-NER) and Biomedical Entity Linking (Bio-EL) are critical for extracting and contextualizing medical information from unstructured texts, such as clinical notes and research articles. Bio-NER focuses on identifying and classifying biomedical entities, including diseases, medications, and genes, while Bio-EL maps these entities to standardized ontologies, like the Unified Medical Language System (UMLS), to ensure consistency and enhance data interoperability. This integration facilitates advanced data analytics and supports various applications, from clinical decision support to biomedical research. Despite significant advancements, challenges persist in handling the inherent variability of medical terminology and disambiguating entities with similar surface forms but different meanings. Recent research efforts are concentrated on improving the accuracy of these processes through sophisticated machine learning models and comprehensive ontologies. In this study, we have employed various Long Short-Term Memory (LSTM) models to perform Named Entity Recognition (NER) on i2b2 datasets from 2010, 2012, and 2018. Our approach involves a comparative analysis of the accuracy of these LSTM models in extracting and classifying biomedical entities from the datasets. Following the recognition phase, we have linked the identified entities to the UMLS database to assess the effectiveness of the entity linking process. The study evaluates the performance of these LSTM models in terms of both entity recognition and linking accuracy, providing insights into their effectiveness for biomedical information processing. This work aims to enhance the reliability and utility of Bio-NER and Bio-EL systems in extracting and standardizing biomedical data.

Keywords: Biomedical Named Entity Recognition (Bio-NER), Biomedical Entity Linking (Bio-EL), Unified Medical Language System (UMLS), Long Short-Term Memory (LSTM), i2b2 datasets

CHAPTER 1: INTRODUCTION

Biomedical Named Entity Recognition (Bio-NER) is a pivotal task in natural language processing (NLP) aimed at identifying and classifying entities within biomedical texts. These entities encompass a wide range of concepts, including diseases, medications, genes, and proteins. Accurate extraction of these entities is essential for various applications, such as clinical decision support systems, health information retrieval, and biomedical research. Bio-NER serves as a foundational component in transforming unstructured medical data into structured information that can be analyzed and utilized effectively.

The task of Bio-NER is complicated by the inherent variability and complexity of biomedical language. Medical texts often contain specialized terminology that can vary in form and meaning across different contexts. For instance, a term like 'heart disease' might refer to different medical conditions or stages of disease depending on its usage. This variability can lead to challenges in accurately recognizing and categorizing entities. Furthermore, the presence of synonyms, acronyms, and abbreviations in biomedical literature adds another layer of complexity, necessitating advanced NLP techniques to achieve reliable entity recognition.

Entity linking, or biomedical entity linking (Bio-EL), extends beyond the recognition of entities to map them to standardized ontologies, such as the Unified Medical Language System (UMLS). UMLS provides a comprehensive and unified framework for representing biomedical concepts and their interrelationships. By linking recognized entities to UMLS, Bio-EL enhances the consistency and interoperability of biomedical data. This mapping process ensures that entities are not only identified but also contextualized within a broader knowledge framework, facilitating more accurate and meaningful data integration and retrieval.

The integration of Bio-NER and Bio-EL with UMLS has significant implications for the management and analysis of biomedical information. By aligning extracted entities with a standardized ontology, researchers and clinicians can achieve a coherent understanding of the data, enabling more effective cross-referencing and comparison. This alignment also supports advanced analytical tasks, such as trend analysis and outcome prediction, by providing a structured and semantically enriched view of the biomedical domain.

Recent advancements in Bio-NER and Bio-EL have focused on addressing challenges related to entity disambiguation and context integration. Innovative approaches, including machine learning models and deep learning techniques, are being developed to improve the accuracy of entity recognition and linking. These models leverage large-scale biomedical corpora and sophisticated algorithms to enhance performance, tackle ambiguities, and adapt to the dynamic nature of medical terminology.

The ongoing research and development in Bio-NER and Bio-EL are crucial for advancing biomedical data processing and utilization. By refining these methods and incorporating comprehensive ontologies like UMLS, the field can achieve more accurate, consistent, and actionable insights from biomedical texts. This progress not only benefits clinical and research applications but also contributes to the broader goal of improving healthcare outcomes and advancing scientific knowledge through enhanced data integration and analysis.

CHAPTER 2: LITERATURE SURVEY

Numerous works have been undertaken over time in the field of Named Entity Recognition and Entity Linking. Here, a variety of approaches for clinical NER and Entity Linking with UMLS by various researchers have been reviewed.

Le et al. in [1] proposed an Active Learning (AL) method for clinical Named Entity Recognition (NER) that improves efficiency by selecting training instances based on the similarity between train and test data distributions. It significantly reduced the amount of training data needed, outperforming baseline methods, especially in early iterations, with notable gains using GloVe and BioBERT embeddings.

Nath et al. in [2] introduced three BiLSTM-CRF-based architectures for Named Entity Recognition (NER) in clinical text, modeling it as a supervised, multi-label tagging problem to capture entities and their attributes. The proposed methods demonstrated strong performance on i2b2/VA datasets, achieving high F1-scores for both entity recognition and attribute detection.

In the work of Narayanan et al. [3], a novel multi-task adaptation method using BERT for Named Entity Recognition (NER) of medication and adverse drug events (ADE) in clinical text was proposed, improving performance across diverse clinical tasks. The integrated model outperformed standalone BERT baselines, enhancing both medication extraction and ADE recognition without the need for additional features or new training data.

In the study by Dash et al. [4], a deep neural network model was proposed using pre-trained word embeddings and refined hyperparameters to improve Clinical Named Entity Recognition (Bio-NER) from Electronic Medical Records (EMRs), achieving an F1-Score of 88.34%. The model eliminated the need for post-processing, enhancing both accuracy and efficiency in extracting biomedical entities.

In their research detailed in [5], French et al. surveyed the evolution of Biomedical Entity Linking (BEL), a crucial NLP task for mapping text spans to unique identifiers in ontologies, from its inception to current state-of-the-art systems. It covered available datasets, shared tasks, technical components, and potential future directions for the field.

In their work cited as [6], Nath et al. proposed an unsupervised, interpretable method for automatic clinical coding that bypasses the need for annotated data by leveraging Named

Entity and Attribute Recognition along with clinical domain-specific word embeddings. The approach efficiently extracted and encoded crucial information from clinical notes to facilitate accurate coding.

Vashishth et al. introduced MEDTYPE, a semantic type prediction module that enhanced biomedical NLP pipelines by filtering irrelevant candidate concepts during medical entity linking, improving performance across various toolkits and datasets in [7]. Additionally, they presented two large-scale datasets, WIKIMED and PUBMEDDS, for training, achieving state-of-the-art results in semantic type prediction for biomedical text.

In [8], Kartchner et al. systematically evaluated nine state-of-the-art biomedical entity linking (Bio-EL) models, comparing them on accuracy, speed, ease of use, generalization, and adaptability. The evaluation identified significant gaps, particularly in linking genes and proteins, and highlights the challenge of incorporating context in linking decisions, with the evaluation framework and models made publicly available on GitHub.

In their work referenced as [9], Chanda et al. introduced *definition2vec*, a novel algorithm that enhances the learning of embeddings for medical terms from limited medical notes by incorporating definitions from the UMLS Metathesaurus. The approach effectively grouped semantically similar terms in the embedding space, even for rare or unseen terms, and proved useful in downstream medical informatics applications.

In [10], Lin et al. presented kNN-BioEL, a new biomedical entity linking approach that enhances generalization by referencing similar instances from the training corpus and employing a contrastive learning objective with dynamic hard negative sampling. The method outperformed state-of-the-art baselines, particularly in handling rare and difficult entities across multiple datasets.

Zhu et al. introduced BioPRO, a two-stage biomedical entity linking algorithm that improved representation by combining coarse-grained retrieval with context-aware prompt learning, in [11]. The method addressed both variety and ambiguity challenges, achieving significant performance gains over state-of-the-art techniques across multiple biomedical datasets.

CHAPTER 3: OBJECTIVES

The objective of the study is to improve the performance of biomedical entity recognition from i2b2 medical records and linking the entities with UMLS (Unified Medical Language System), which include the following goals:-

i. To extract information about all of the medication entity that are known to be taken by the patient, for each patient medical record provided, i.e. annotation of all i2b2 2010, 2012 and 2018 datasets with medical entities.
ii. To check and compare the performance of Named Entity Recognition (NER) of various LSTM-based models on all annotated datasets of first task
iii. To link the entities identified in first task with UMLS, i.e. to embed Ontological Knowledge in Biomedical Entity Linking within medical records.

The ultimate objective of this assessment is to enhance biomedical entity recognition and linking in i2b2 medical records by annotating medication entities across the i2b2 2010, 2012 and 2018 datasets. It evaluates and compares the performance of Named Entity Recognition (NER) of various LSTM-based models on these annotated datasets, comparing it with previous studies. Additionally, the study focuses on linking these identified entities to the Unified Medical Language System (UMLS), integrating ontological knowledge into Biomedical Entity Linking to improve the identification and standardization of biomedical mentions such as diseases and drugs within medical records.

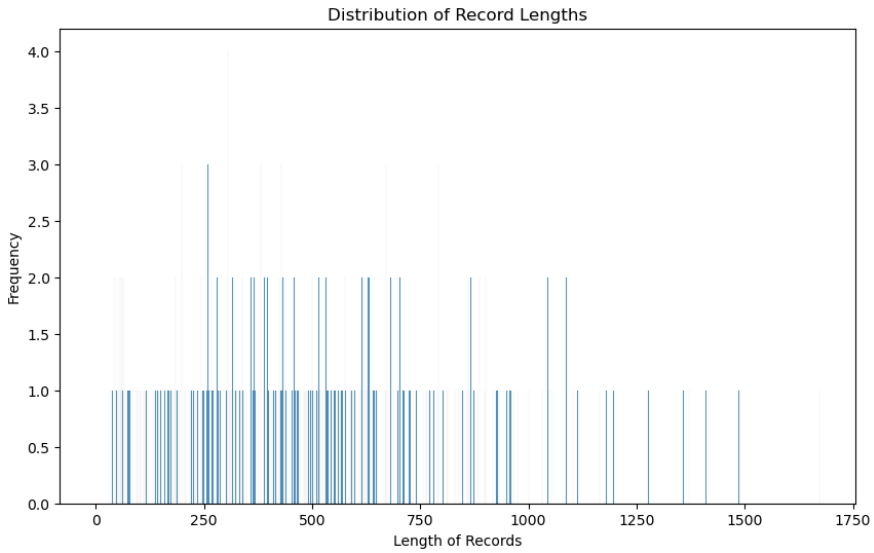
CHAPTER 4: METHODOLOGY & IMPLEMENTATION

Dataset Description

Three i2b2 datasets have been used for this study. These datasets are primarily derived from de-identified clinical records contributed by healthcare institutions that participate in the i2b2 (Informatics for Integrating Biology and the Bedside) challenges.

1. <u>i2b2 2010</u>

   This dataset consists of de-identified discharge summaries and progress notes from patients. It contains annotations for medical concepts like problems, treatments, and tests, as well as the relations between them, such as whether a treatment improves or worsens a problem. It contains a total of 3,74,397 entities.



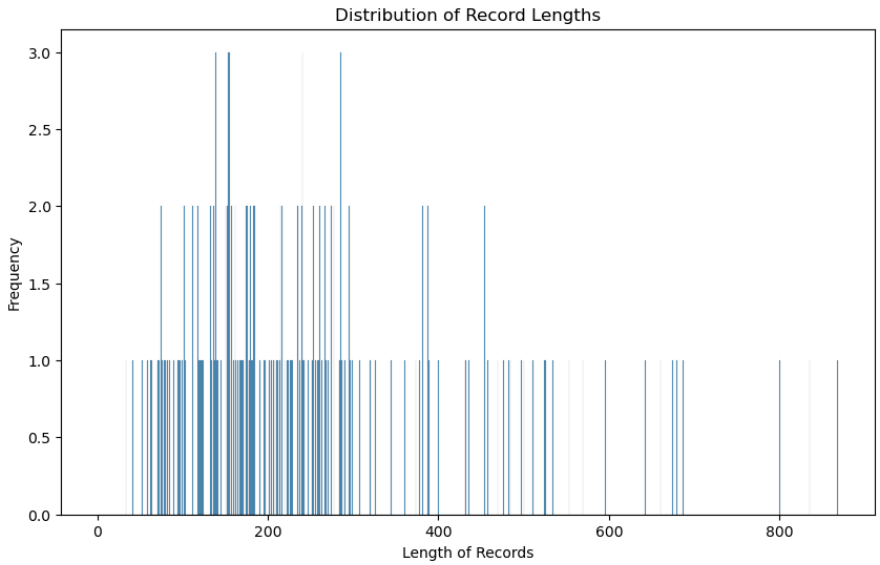
O	98143
I-problem	19581
B-problem	13110
I-treatment	8745
I-test	8737
B-treatment	8335
B-test	7188

(a) *(b)*

Fig. 4.1 – i2b2 2010 dataset: (a) Distribution of Record Lengths, (b) Entity Count for different Labels

2. <u>i2b2 2012</u>

   This dataset includes clinical notes annotated for medical events (e.g., symptoms, diagnoses, treatments), time expressions (e.g., "on Monday", "in 2019"), and temporal relations (e.g., an event happening before or after another). It contains a total of 1,03,502 entities.



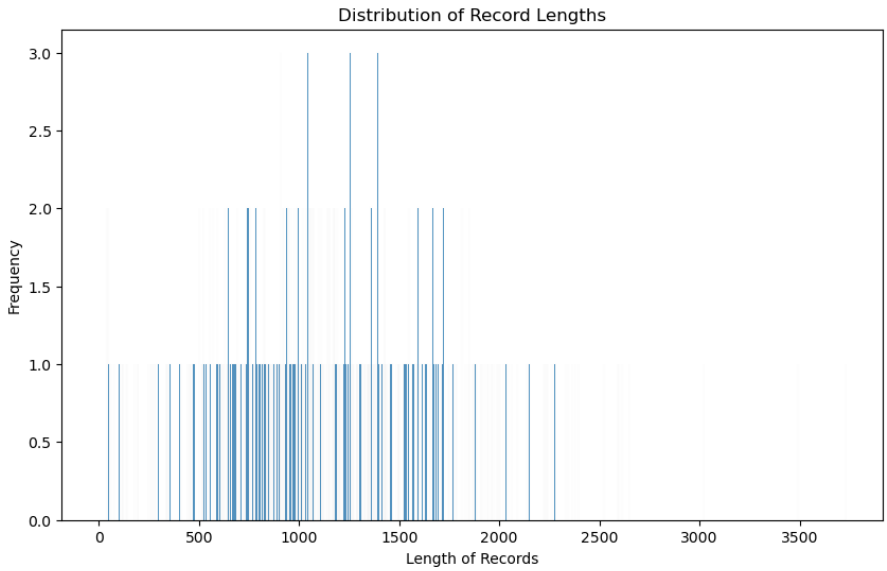
O	18271
I-PROBLEM	6200
B-PROBLEM	3631
I-TREATMENT	3116
I-TEST	2872
B-OCCURRENCE	2821
B-TREATMENT	2464
I-OCCURRENCE	1537
I-CLINICAL_DEPT	1368
B-TEST	1269
I-DATE	1069
B-EVIDENTIAL	713
I-DURATION	532
B-DATE	388
B-DURATION	282
B-FREQUENCY	222
B-CLINICAL_DEPT	173
I-FREQUENCY	148
I-EVIDENTIAL	61
I-TIME	48
B-TIME	17

(a) *(b)*

Fig. 4.2 – i2b2 2012 dataset: (a) Distribution of Record Lengths, (b) Entity Count for different Labels

3. <u>i2b2 2018</u>

This dataset contains de-identified patient records annotated for mentions of medications, dosage, frequency, mode of administration, and ADEs. It also includes labels for whether an ADE is associated with a specific medication. It contains a total of 9,43,987 entities.



O	465570
B-Drug	23058
B-Form	11323
B-Strength	9257
B-Route	8828
B-Reason	8533
B-Frequency	7758
I-Strength	7158
I-Frequency	6022
I-Dosage	5023
B-Dosage	4825
I-Reason	3740
I-Form	2992
I-Drug	2705
B-ADE	1866
I-Duration	1323
I-ADE	883
I-Route	550
B-Duration	464

(a) *(b)*

Fig. 4.3 – i2b2 2018 dataset: (a) Distribution of Record Lengths, (b) Entity Count for different Labels

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Methodology & Implementation

(I) Named Entity Recognition (NER)

For the task of Named Entity Recognition (NER) in clinical text, effectively identifying key medical entities such as diseases, treatments, medications, and adverse drug events (ADEs) is crucial for enhancing healthcare systems. The clinical narratives from the 2010, 2012, and 2018 challenges comprise the i2b2 datasets, which offer a strong basis for the development of NER models in medical NLP. NLP systems have distinct challenges as a result of the datasets' extensive capture of medical concepts, temporal correlations, and medication-related adverse effects. To tackle these challenges, various deep learning models, including LSTM, BiLSTM, BiLSTM with Time Distributed Dense Layers, and BiLSTM with Time Distributed Dense Layers and CNN, have been employed. To identify clinical entities more accurately, these models use several techniques to extract contextual information and local patterns from text. The effectiveness of these models in processing complicated medical language and enhance automated systems for clinical data extraction has been assessed by evaluating how well they perform on the i2b2 datasets.

A. **Model Implementation**

(i) LSTM (Long Short-Term Memory)

A single-directional LSTM processes the input sequentially and is useful for learning temporal dependencies in sequential data.

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 embedding (Embedding)       (None, 1000, 216)         1924776

 bidirectional (Bidirectiona  (None, 1000, 128)        143872
 l)

 dense (Dense)               (None, 1000, 8)           1032

=================================================================
Total params: 2,069,680
Trainable params: 2,069,680
Non-trainable params: 0
_________________________________________________________________
```

Fig. 4.4 – Model parameters for LSTM implementation

(ii) BiLSTM (Bidirectional LSTM)

A BiLSTM extends the LSTM by processing the input in both forward and backward directions, providing access to both past and future contexts.

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 embedding (Embedding)       (None, 1000, 64)          570304

 bidirectional (Bidirectiona  (None, 1000, 128)        66048
 l)

 bidirectional_1 (Bidirectio  (None, 1000, 64)         41216
 nal)

 dense (Dense)               (None, 1000, 64)          4160

 dense_1 (Dense)             (None, 1000, 8)           520

=================================================================
Total params: 682,248
Trainable params: 682,248
Non-trainable params: 0
_________________________________________________________________
```

Fig. 4.5 – Model parameters for BiLSTM implementation

(iii) BiLSTM with Time Distributed Dense Layers

In this architecture, each timestep's output from the BiLSTM is passed through a dense layer using the TimeDistributed wrapper. This allows each word or token in the sequence to get its own fully connected network output.

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 embedding (Embedding)       (None, 1000, 64)          570304

 bidirectional (Bidirectiona  (None, 1000, 128)        66048
 l)

 time_distributed (TimeDistr  (None, 1000, 64)         8256
 ibuted)

 time_distributed_1 (TimeDis  (None, 1000, 8)          520
 tributed)

=================================================================
Total params: 645,128
Trainable params: 645,128
Non-trainable params: 0
_________________________________________________________________
```

Fig. 4.6 – Model parameters for BiLSTM with Time Distributed Dense Layers implementation

(iv) BiLSTM with Time Distributed Dense Layers and CNN (Convolutional Neural Networks)

This hybrid model combines the BiLSTM with CNN layers before applying Time Distributed Dense Layers. CNNs are used to capture local features or patterns (like prefix or suffix in words) that may be helpful in identifying specific entities.

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 embedding (Embedding)       (None, 1000, 64)          570304

 conv1d (Conv1D)             (None, 1000, 32)          6176

 bidirectional (Bidirectiona  (None, 1000, 128)        49664
 l)

 time_distributed (TimeDistr  (None, 1000, 64)         8256
 ibuted)

 dropout (Dropout)           (None, 1000, 64)          0

 dense_1 (Dense)             (None, 1000, 8)           520

=================================================================
Total params: 634,920
Trainable params: 634,920
Non-trainable params: 0
_________________________________________________________________
```

Fig. 4.7 – Model parameters for BiLSTM with Time Distributed Dense Layers and CNN implementation

B. **Model Evaluation**

The *confusion matrix* and *classification report*, containing statistics like *precision*, *recall*, *f1-score* and *support* for each class, together with their *macro average*, *weighted average* and the overall *accuracy*, are generated for each model.

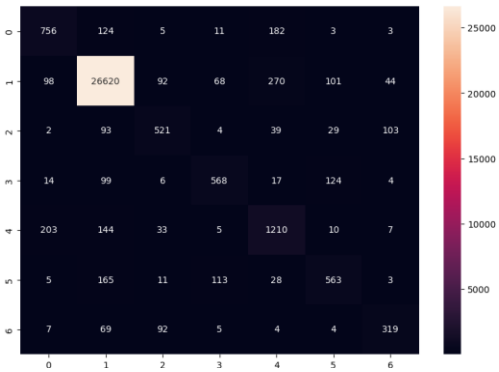
The descriptions of few evaluation criteria are as follows:

- *Confusion Matrix*: A confusion matrix presents a table layout of the different outcomes of the prediction and results of a classification problem, and help visualize its outcomes.
- *True Positive (TP):* The number of times the actual positive values are equal to the predicted positive values.
- *False Positive (FP):* The number of times the model wrongly predicts negative values as positives.
- *True Negative (TN):* The number of times the actual negative values are equal to the predicted negative values.

- *False Negative (FN):* The number of times the model wrongly predicts positive values as negatives.
- *Accuracy:* $$\frac{\text{No. of Correct Predictions}}{\text{Total No. of Predictions}} = \frac{TP + TN}{TP + FP + TN + FN} \quad \textbf{(4.1)}$$
- *Precision:* $$\frac{TP}{TP + FP} \quad \textbf{(4.2)}$$
- *Recall:* $$\frac{TP}{TP + FN} \quad \textbf{(4.3)}$$
- *f1-Score:* $$\frac{2 * Precision * Recall}{(Precision + Recall)} \quad \textbf{(4.4)}$$

Comparative Study:-

The four models were compared on the basis of *Accuracy*, *Precision*, *Recall* and *f1-Score*, demonstrated through *confusion matrix* and *classification report*. The metrics for i2b2 2010 dataset for the four models are illustrated in Fig. 4.8-4.11.

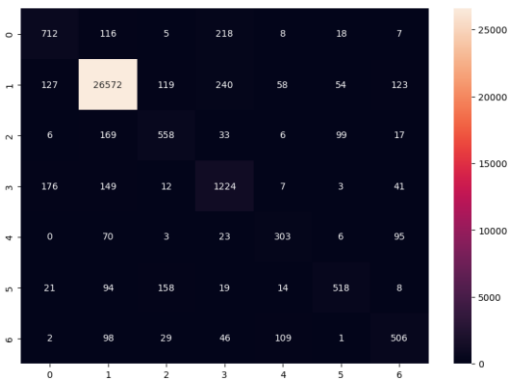


(a)

	precision	recall	f1-score	support
0	0.70	0.70	0.70	1084
1	0.97	0.98	0.97	27293
2	0.69	0.66	0.67	791
3	0.73	0.68	0.71	832
4	0.69	0.75	0.72	1612
5	0.68	0.63	0.65	888
6	0.66	0.64	0.65	500
accuracy			0.93	33000
macro avg	0.73	0.72	0.72	33000
weighted avg	0.93	0.93	0.93	33000

(b)

Fig. 4.8 – LSTM implementation on i2b2 2010 dataset: (a) Confusion Matrix, (b) Classification Report

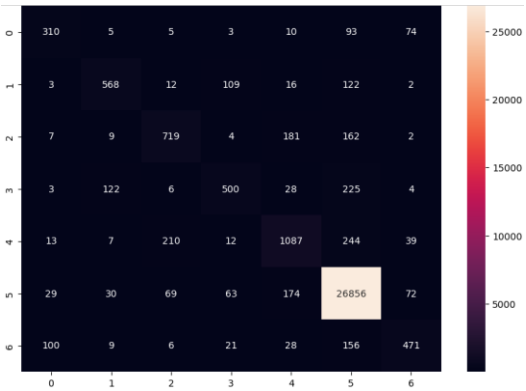


(a)

	precision	recall	f1-score	support
0	0.68	0.66	0.67	1084
1	0.97	0.97	0.97	27293
2	0.63	0.63	0.63	888
3	0.68	0.76	0.72	1612
4	0.60	0.61	0.60	500
5	0.74	0.62	0.68	832
6	0.63	0.64	0.64	791
accuracy			0.92	33000
macro avg	0.71	0.70	0.70	33000
weighted avg	0.92	0.92	0.92	33000

(b)

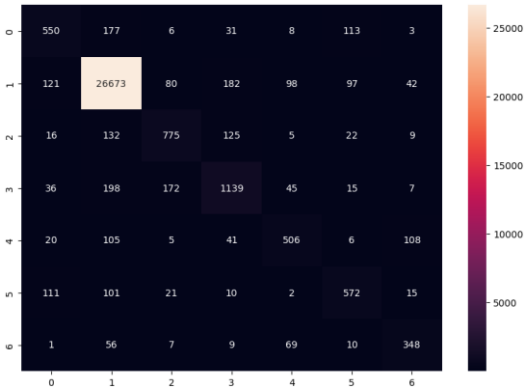
Fig. 4.9 – BiLSTM implementation on i2b2 2010 dataset: (a) Confusion Matrix, (b) Classification Report



	precision	recall	f1-score	support
0	0.67	0.62	0.64	500
1	0.76	0.68	0.72	832
2	0.70	0.66	0.68	1084
3	0.70	0.56	0.63	888
4	0.71	0.67	0.69	1612
5	0.96	0.98	0.97	27293
6	0.71	0.60	0.65	791
accuracy			0.92	33000
macro avg	0.74	0.68	0.71	33000
weighted avg	0.92	0.92	0.92	33000

(a) *(b)*

Fig. 4.10 – BiLSTM with Time Distributed Dense Layers implementation on i2b2 2010 dataset: (a) Confusion Matrix, (b) Classification Report



	precision	recall	f1-score	support
0	0.64	0.62	0.63	888
1	0.97	0.98	0.97	27293
2	0.73	0.71	0.72	1084
3	0.74	0.71	0.72	1612
4	0.69	0.64	0.66	791
5	0.69	0.69	0.69	832
6	0.65	0.70	0.67	500
accuracy			0.93	33000
macro avg	0.73	0.72	0.72	33000
weighted avg	0.93	0.93	0.93	33000

Fig. 4.11 – BiLSTM with Time Distributed Dense Layers and CNN implementation on i2b2 2010 dataset: (a) Confusion Matrix, (b) Classification Report

(II) <u>Entity Linking with UMLS</u>

Entity Linking in the context of clinical text involves mapping identified medical entities (such as diseases, drugs, or treatments) to standardized concepts in knowledge bases like the Unified Medical Language System (UMLS). Extensive resources for Named Entity Recognition (NER), with an emphasis on medical concept extraction, temporal linkages, and adverse medication events, are available in the i2b2 datasets from the 2010, 2012, and 2018 challenges. Recognizing entities, however, is only the first step. Semantic interoperability, which is essential for applications like clinical decision support, patient safety monitoring, and medical research, is ensured by connecting these entities to UMLS concepts. The integration of clinical data across systems can be improved by unifying disparate terminologies by connecting the extracted items to their matching UMLS identifiers. In addition to improving the usability of NER outputs, this procedure helps to strengthen

medical NLP systems that can convert unstructured text into organized, useful medical knowledge.

A. **Implementation & Evaluation**

(i) Extracting Entities from i2b2 Datasets

- Named Entity Recognition (NER) is first performed on the i2b2 datasets (2010, 2012, 2018) to identify medical entities such as diseases, treatments, medications, or temporal expressions. These entities are extracted from clinical notes based on predefined categories (e.g., problem, treatment, test, ADE).
- Example extracted entities: "diabetes," "aspirin," "CT scan."

(ii) Retrieving UMLS Concepts

- UMLS (Unified Medical Language System) is a comprehensive biomedical vocabulary that provides standardized terms across various medical terminologies. For entity linking, the relevant UMLS concepts corresponding to each entity category (e.g., diseases, treatments, medications) are retrieved.
- UMLS concepts are structured with Concept Unique Identifiers (CUIs), which are used to represent a single concept across multiple synonyms and terminologies.
- Example UMLS entities: "Diabetes Mellitus" (CUI: C0011849), "Aspirin" (CUI: C0004057), "CT Scan" (CUI: C0033144).

(iii) Generating BERT Embeddings

- Both i2b2 entities and UMLS concepts are converted into dense vector representations using BERT embeddings. BERT (Bidirectional Encoder Representations from Transformers) is used because it provides contextualized embeddings that capture the semantics of each entity.
    - For the i2b2 entity: The model takes the entity as input and generates an embedding that captures its semantic meaning in the context of the clinical text.
    - For the UMLS entity: Similarly, the UMLS concept (and potentially its synonyms) is input into the model, generating a corresponding embedding.

```python
tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
model = BertModel.from_pretrained('bert-base-uncased')

def get_embedding(text):
    inputs = tokenizer(text, return_tensors='pt', truncation=True, padding=True, max_length=10)
    outputs = model(**inputs)
    return outputs.last_hidden_state.mean(dim=1).detach().numpy()
```

Fig. 4.12 – Code snippet of the function for BERT embedding generation

(iv) Cosine Similarity Computation

- Cosine similarity is then computed between the BERT embeddings of the i2b2 entity and each UMLS entity. Cosine similarity measures the cosine of the angle between two vectors in a multi-dimensional space, providing a score between -1 (completely dissimilar) and 1 (exact match).
- Formula for cosine similarity:

$$\text{Cosine Similarity} = \frac{\|A\|\|B\|}{A \cdot B} \quad \textbf{(4.5)}$$

Where A is the i2b2 entity embedding, and B is the UMLS entity embedding.

(v) Applying a Matching Threshold

- The similarity score obtained from cosine similarity is then compared to a predefined threshold. In this case, a threshold of 85% (0.85) is used.
    - If the cosine similarity score is greater than or equal to 0.85, it is considered a match between the i2b2 entity and the UMLS entity.
    - If the score is less than 0.85, it is marked as no match.
- This threshold helps in determining which UMLS concept is semantically closest to the i2b2 entity while minimizing false matches.

```python
from sklearn.metrics.pairwise import cosine_similarity
def find_matches(entities, entity_embeddings, second_entity_embeddings, threshold=0.85):
    matches = {}
    for entity, embedding in entity_embeddings.items():
        max_similarity = 0
        best_match = None
        for second_entity, second_embedding in second_entity_embeddings.items():
            similarity = cosine_similarity(embedding, second_embedding)[0][0]
            if similarity > max_similarity:
                max_similarity = similarity
                best_match = second_entity
        matches[entity] = (max_similarity >= threshold, best_match, max_similarity)
    return matches
```

Fig. 4.13 – Code snippet of the function for Cosine Similarity computation

(vi) Saving Results in a DataFrame

- The results of the entity linking process are saved in a dataframe with the following columns:
    a) **Entity:** The original entity from the i2b2 dataset (e.g., "diabetes").
    b) **Match:** A binary value or flag indicating whether the entity was considered a match (TRUE for match, FALSE for no match).
    c) **Matched Entity:** The UMLS entity that matched based on the cosine similarity score (e.g., "Diabetes Mellitus").
    d) **Score:** The cosine similarity score between the i2b2 entity and the UMLS entity (e.g., 0.87).

	A	B	C	D
1	Entity	Match	Matched Entity	Score
2	Nsaids/Anti-Inflammatory Classifier	FALSE	Adverse effect of antineoplastic drug	0.8310431
3	Mesenteric angiograpm	FALSE	Polyarticular postdysenteric arthritis	0.8435653
4	coil embolization	FALSE	Atrial appendage isomerism	0.7661884
5	bleeding vessel	TRUE	Bleeding	0.89125
6	long term anti-coagulation	FALSE	Long term isotretinoin therapy	0.8438658

Fig. 4.14 – A part of the saved dataframe for i2b2 2010 dataset

CHAPTER 5: RESULT DISCUSSION

(I) Named Entity Recognition

On comparing the four models, the performance varied for the different datasets. Considering the average *Accuracy* for the three datasets, the order is:

BiLSTM (89.33%) < *BiLSTM with Time Distributed Dense Layers and CNN* (90%) < *BiLSTM with Time Distributed Dense Layers* (90.33%) < *LSTM (91%)*

The statistics are demonstrated in Table 5.1.

Table 5.1: Accuracy, Precision, Recall and f1-Score Percentages for the four Models implemented on the three i2b2 datasets

	i2b2_2010	i2b2_2012	i2b2_2018
LSTM	Accuracy: 93%	Accuracy: 83%	Accuracy: 97%
	Precision: 93%	Precision: 82%	Precision: 96%
	Recall: 93%	Recall: 83%	Recall: 97%
	f1-Score: 93%	f1-Score: 82%	f1-Score: 96%
BiLSTM	Accuracy: 92%	Accuracy: 81%	Accuracy: 95%
	Precision: 92%	Precision: 80%	Precision: 93%
	Recall: 92%	Recall: 81%	Recall: 95%
	f1-Score: 92%	f1-Score: 80%	f1-Score: 94%
BiLSTM with Time Distributed Dense Layers	Accuracy: 92%	Accuracy: 83%	Accuracy: 96%
	Precision: 92%	Precision: 81%	Precision: 95%
	Recall: 92%	Recall: 83%	Recall: 96%
	f1-Score: 92%	f1-Score: 82%	f1-Score: 95%
BiLSTM with Time Distributed Dense Layers and CNN	Accuracy: 93%	Accuracy: 81%	Accuracy: 96%
	Precision: 93%	Precision: 80%	Precision: 95%
	Recall: 93%	Recall: 81%	Recall: 96%
	f1-Score: 93%	f1-Score: 81%	f1-Score: 95%

(II) Entity Linking with UMLS

The performance of Entity Linking with UMLS was evaluated based on the percentage of entities matched. This evaluation is crucial for understanding how effectively the system is linking extracted entities from the i2b2 datasets to their corresponding concepts in UMLS.

For evaluating the performance of linking on the three datasets, the percentage of matched entities is calculated by dividing the number of successfully matched entities by the total number of extracted entities, and multiplying by 100 to get the percentage.

Formula:-

$$\textit{Percentage of Matched Entities} = \frac{\textit{Number of Matched Entities}}{\textit{Total Entities Extracted}} \times 100 \qquad \textbf{(5.1)}$$

The statistics are demonstrated in Table 5.2 and Table 5.3

Table 5.2: Percentage of Matched Entities for the three i2b2 datasets

i2b2_2010	i2b2_2012	i2b2_2018
29.94%	22.61%	46.56%

Table 5.3: Percentage of Matched Entities for the implementation of three versions of BERT on i2b2 2018 dataset

BERT	BioBERT	SciBERT
46.56%	99.05%	47.35%

CHAPTER 6: CONCLUSION & FUTURE SCOPE

The combination of Named Entity Recognition (NER) and Entity Linking (EL) plays a vital role in processing clinical text, allowing us to extract and map important medical terms like diseases, medications, and treatments to standard concepts in knowledge bases such as the UMLS. This is necessary to convert unstructured clinical notes into structured data, which will facilitate patient data administration, enhance healthcare decision-making, and promote medical research.

The accuracy of detecting and linking medical entities has greatly increased with the use of deep learning models like LSTM and BiLSTM and sophisticated methods like BERT embeddings. Nonetheless, there are still issues to be resolved, like managing unclear or uncommon terminology and making sure all identified entities are accurately connected to UMLS concepts. The evaluation, based on the percentage of successfully matched entities, helps measure the effectiveness of these systems. It also demonstrated the efficacy of BioBERT embeddings in the implementation of clinical (medical) entities linking.

The future of Named Entity Recognition (NER) and Entity Linking (EL) in clinical text holds great promise, with opportunities for improving model accuracy through domain-specific fine-tuning and expanding knowledge bases like UMLS to include more medical terminology. Additionally, incorporating multilingual capabilities can enhance accessibility across diverse healthcare settings. Ultimately, integrating these advanced systems into electronic health records (EHR) will streamline clinical workflows, enabling faster and more accurate decision-making in patient care.

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